

Amazon Consumer Behavior Analysis

By

Raj Verma

Introduction

Background

E-commerce, short for electronic commerce, is the purchase and sale of products and services via the internet and online platforms. Its beginnings can be traced back to the 1960s, when electronic data interchange (EDI) enabled organizations to exchange documents and conduct transactions electronically. However, the modern idea of e-commerce began to take shape in the 1990s, with the rise of the internet and technological advancements.

Amazon has always been interconnected with the evolution and expansion of e-commerce. As one of the field's pioneers and most important firms, Amazon has had a huge impact on the landscape of online retail and e-commerce in general. Founded in 1994 by Jeff Bezos, Amazon began as an online bookstore before expanding into multiple product categories and becoming the e-commerce behemoth we know today. Bezos chose to start with books due to their widespread appeal and extensive selection.

Since its inception, Amazon has prioritized the consumer experience, implementing features such as customer reviews and personalized suggestions. Its expansion into electronics, apparel, and household goods has allowed Amazon to reach a larger customer base. To support its growing activities, the company made significant infrastructure investments, including fulfillment centers and logistics systems. Innovations like one-click ordering and Amazon Prime membership have further enhanced the buying experience.

Strategic acquisitions, such as Zappos, and partnerships have played a crucial role in Amazon's growth. Additionally, international expansion has strengthened its global presence.

Objectives

This information was produced to analyze the behavioral patterns of Amazon's customers. It comprises a wide range of customer interactions and browsing habits across the Amazon ecosystem.

It incorporates various factors such as customer demographics, user engagement, and reviews. The dataset aims to provide insight into Amazon customers' preferences, purchasing behaviors, and decision-making processes. By analyzing this dataset, researchers and analysts can gain a better understanding of consumer behavior, identify patterns, optimize marketing strategies, and enhance the overall Amazon customer experience.

This project analyzes customer behavior and the correlations between features in the dataset.

Research Question

What are the key factors influencing Amazon customers' preferences and purchasing behaviors, and how do these factors vary across different demographics?

Rationale

Why this question?

This research subject was chosen based on its practical significance and potential impact. Understanding the factors that influence Amazon customers' preferences and purchasing behaviors is crucial for companies aiming to succeed in the e-commerce landscape.

What drew us to it?

The interest lies in breaking down the complex web of customer decision-making. Identifying these factors empowers organizations to adjust their strategies, enhance customer experiences, and optimize resource allocation.

Worth exploring?

This question merits consideration as it bridges the gap between theory and practice. It enables us to uncover actionable insights that can directly influence company outcomes.

Relevance

Significance in today's business landscape

Customer-Centric Approach: In an era where customer-centricity is paramount, understanding customer behavior is crucial. Businesses that grasp their customers' desires, preferences, and pain points can provide tailored solutions.

Market Competition: The e-commerce market is highly competitive, and Amazon, as a global powerhouse, faces intense competition. Understanding customer behavior can confer a competitive advantage. Efficient resource allocation relies on knowing where to invest. Understanding customer preferences enables Amazon to allocate resources more effectively.

Personalization: Personalized marketing and product recommendations are essential. Understanding what motivates customers enables targeted interactions.

Market Trends: The e-commerce landscape is evolving rapidly. The findings of this study can help Amazon strategize its initiatives in this dynamic industry.

Why should anyone care?

Business Stakeholders: Executives, marketing, and product managers require these responses. They make commercial judgments.

Investors: Understanding customer behavior has an impact on stock performance. Investors are interested in learning more about Amazon's growth prospects.

Consumers: Better customer service helps everyone. Better recommendations, quicker transactions, and relevant content all increase customer satisfaction.

Data Analysis

Importing Python libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from plotly.subplots import make_subplots
import plotly.graph_objects as go
import scipy
from scipy.stats import spearmanr
import statistics
from datetime import timedelta
import warnings
warnings.simplefilter("ignore")
```

Loading Data

```
data = pd.read_csv('Amazon.csv')
data.head()
```

Timestamp	age	Gender	Purchase_Frequency	Purchase_Categories	Personalized_Recommendation_Frequency
0	2023/06/04 1:28:19 PM GMT+5:30		23 Female	Few times a month	Beauty and Personal Care
1	2023/06/04 2:30:44 PM GMT+5:30		23 Female	Once a month	Clothing and Fashion
2	2023/06/04 5:04:56 PM GMT+5:30		24 Prefer not to say	Few times a month	Groceries and Gourmet Food;Clothing and Fashion
3	2023/06/04 5:13:00 PM GMT+5:30		24 Female	Once a month	Beauty and Personal Care;Clothing and Fashion;...
4	2023/06/04 5:28:06 PM GMT+5:30		22 Female	Less than once a month	Beauty and Personal Care;Clothing and Fashion

Data Description

The dataset used aims to offer insights into customer preferences, shopping tendencies, and decision-making mechanisms on the Amazon platform.

Sr. No.	Feature name	Feature description
1	Timestamp	Date/Time
2	age	age
3	Gender	gender
4	Purchase_Frequency	How frequently do you make purchases on Amazon?
5	Purchase_Categories	What product categories do you typically purchase on Amazon?
6	Personalized_Recommendation_Frequency	Have you ever made a purchase based on personalized product recommendations from Amazon?
7	Browsing_Frequency	How often do you browse Amazon's website or app?
8	Product_Search_Method	How do you search for products on Amazon?
9	Search_Result_Exploration	Do you tend to explore multiple pages of search results or focus on the first page?
10	Customer_Reviews_Importance	How important are customer reviews in your decision-making process?
11	Add_to_Cart_Browsing	Do you add products to your cart while browsing on Amazon?
12	Cart_Completion_Frequency	How often do you complete the purchase after adding products to your cart?
13	Cart_Abandonment_Factors	What factors influence your decision to abandon a purchase in your cart?
14	Saveforlater_Frequency	Do you use Amazon's "Save for Later" feature, and if so, how often?
15	Review_Left	Have you ever left a product review on Amazon?
16	Review_Reliability	How much do you rely on product reviews when making a purchase?
17	Review_Helpfulness	Do you find helpful information from other customers' reviews?
18	Personalized_Recommendation_Frequency	How often do you receive personalized product recommendations from Amazon?
19	Recommendation_Helpfulness	Do you find the recommendations helpful?
20	Rating_Accuracy	How would you rate the relevance and accuracy of the recommendations you receive
21	Shopping_Satisfaction	How satisfied are you with your overall shopping experience on Amazon?
22	Service_Appreciation	What aspects of Amazon's services do you appreciate the most?
23	Improvement_Areas	Are there any areas where you think Amazon can improve?

```
print("The data size -->",data.shape)
```

Output :- The data size --> (602, 23)

This line prints the shape of the data, which is represented as a tuple (number of rows, number of columns). In this case, it shows there are 602 rows and 23 columns in the data.

```
data.describe()
```

	age	Customer_Reviews_Import ance	Personalized_Recommendation_Freq uency	Rating_Accuracy	Shopping_Satisfaction
count	602	602	602	602	602
mean	30.790698	2.480066	2.699336	2.672757	2.463455
std	10.193276	1.185226	1.042028	0.899744	1.012152
min	3	1	1	1	1
25%	23	1	2	2	2
50%	26	3	3	3	2
75%	36	3	3	3	3
max	67	5	5	5	5

Interpretation of the data :-

1. **Age:** Describes the age distribution of the customers surveyed, ranging from 3 to 67 years old. The mean age is approximately 30.79 years, with a standard deviation of approximately 10.19 years.
2. **Customer_Reviews_Importance:** Indicates the perceived importance of customer reviews to the surveyed customers, rated on a scale from 1 to 5. The mean rating is approximately 2.48, with a standard deviation of approximately 1.19.
3. **Personalized_Recommendation_Frequency:** Reflects the frequency at which personalized recommendations are received by customers, rated on a scale from 1 to 5. The mean rating is approximately 2.70, with a standard deviation of approximately 1.04.
4. **Rating_Accuracy:** Represents the perceived accuracy of product ratings on the platform, rated on a scale from 1 to 5. The mean rating is approximately 2.67, with a standard deviation of approximately 0.90.
5. **Shopping_Satisfaction:** Indicates the overall satisfaction level of customers with their shopping experience, rated on a scale from 1 to 5. The mean rating is approximately 2.46, with a standard deviation of approximately 1.01.

By looking at these descriptive statistics, you can get a quick sense of the data in your dataset. For example, you can see that the customers tend to be somewhat young adults (average

around 30 years old) and that their ratings and satisfaction scores lean towards the positive side (average above 2).

Data Cleaning

→ Rename the first "Personalized_Recommendation_Frequency" column to "Recomendation Purchase Status"

```
data.rename(columns= {"Personalized_Recommendation_Frequency" :  
"Recomendation_Purchase_Status"} , inplace= True)
```

→ Replacing the "Never" value from cart_Completion_Frequency with "Data not available".

```
data.loc[data['Cart_Completion_Frequency'] == "Never" , ['Cart_Completion_Frequency']] =  
'Data not Available'
```

→ Adding Date and Time columns by extracting data from Timestamp column

```
data['Timestamp'] = pd.to_datetime(data['Timestamp'])  
data['Timestamp'] += timedelta(hours=5, minutes=30)  
data['Date'] = data['Timestamp'].dt.strftime('%d/%m/%Y')  
data['Time'] = data['Timestamp'].dt.time
```

→ Adding Age Category Column

```
def age_cat(row):  
    x = row['age']  
    if x < 18 :  
        return 'Minor (<18)'  
    elif (x > 18) & (x <= 25):  
        return 'Young Adult (18-25)'  
    elif (x > 25) & (x <= 40):  
        return "Adult (26-40)"  
    elif (x > 40) & (x <= 60) :  
        return 'Middle-Aged Adult (41-60)'  
    else :  
        return 'Senior Citizen (>60)'  
data['Age_Category'] = data.apply(age_cat , axis = 1)
```

→ Adding Time Status Column

```
def Time_cat(x):  
    if 6 <= x.hour < 12:
```

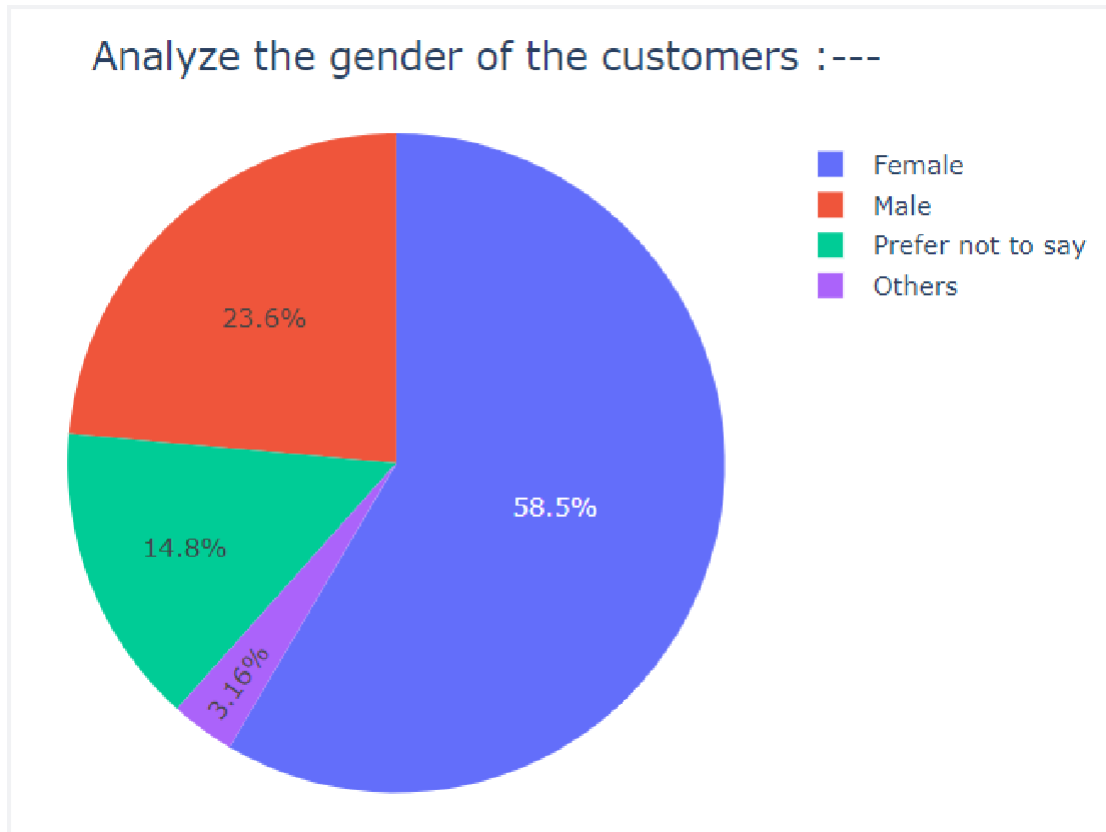


```
    return 'Morning'  
elif 12 <= x.hour < 18:  
    return 'Afternoon'  
elif 18 <= x.hour < 21:  
    return 'Evening'  
else:  
    return 'Night'
```

```
data['Time_Status'] = data['Time'].apply(Time_cat)
```

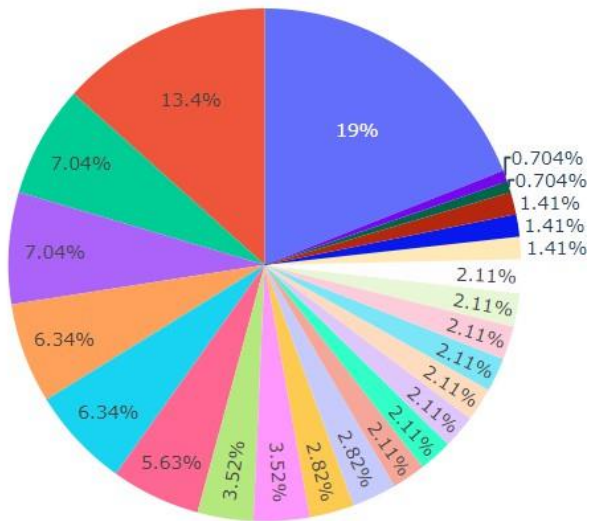
Data Visualization

Gender Analysis of customers

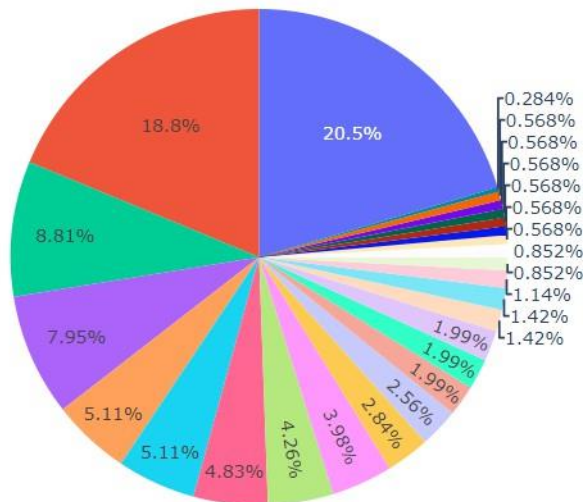


In the pie chart above, the female ratio is more than half, while the male population is less than a quarter. Approximately 15% did not disclose their gender.

Male Purchase Category Distribution

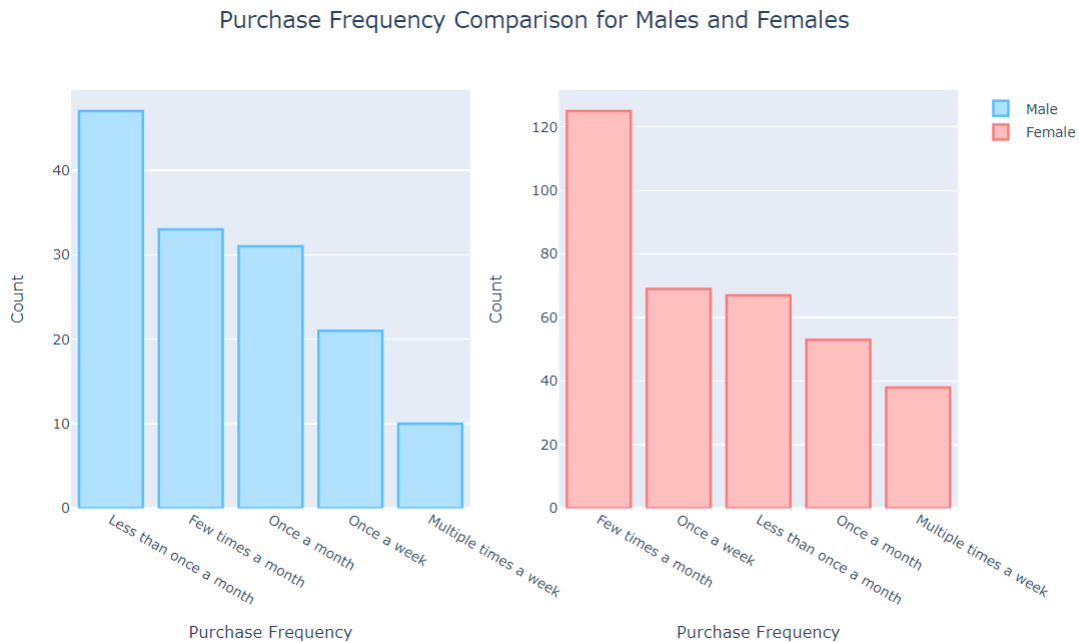


Female Purchase Category Distribution



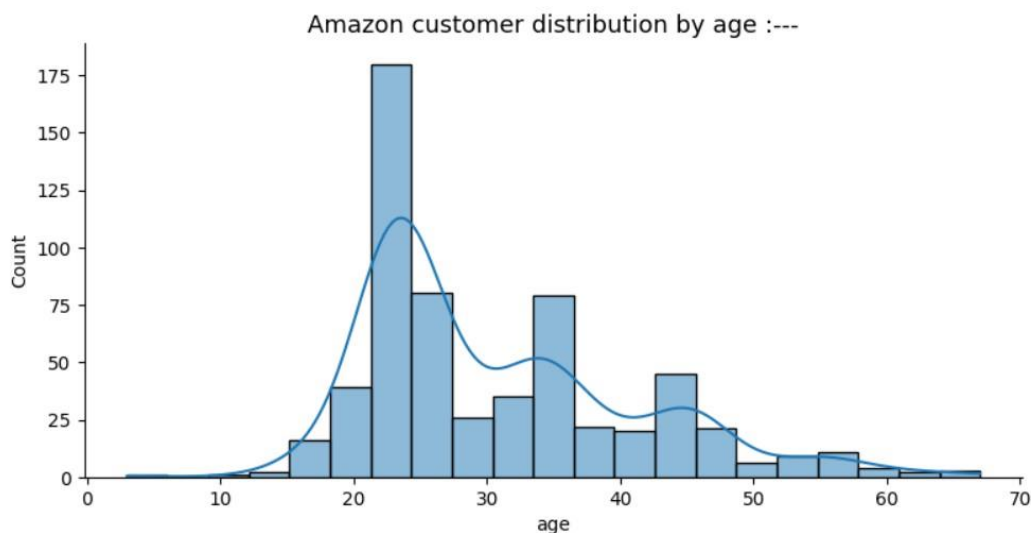
When comparing male and female categorical expenditures, we can infer that both have an equal share in the "other" categories, and "Beauty and Personal Care" along with "Fashion Clothing" have roughly equal shares of 8%. However, the expenditure on Men's Clothing and Fashion is 5% less than that of women, which are 8% and 13%, respectively.

Frequency count of Men and women



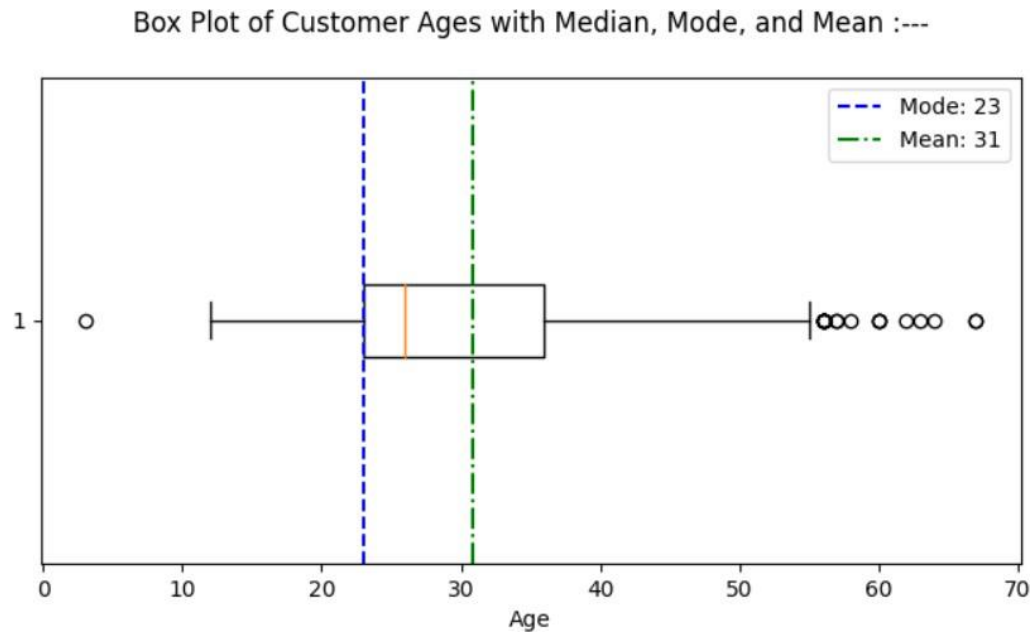
The purchase frequency of men for "few times a month" is a quarter of the female count (120), with a similar trend for "multiple times a week," where the frequencies of males and females are 10 and 40, respectively.

Customer classification based on age



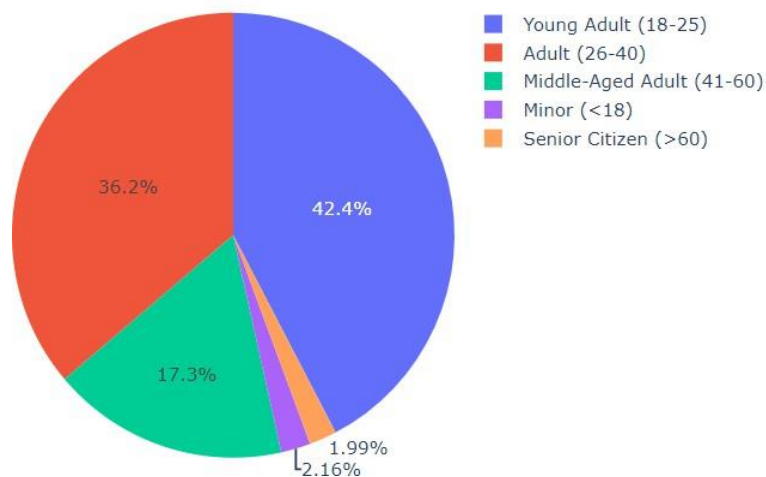
The graph shows an oscillating trend in the count from ages 15 to 25, 25 to 40, and 40 to 50. Additionally, the highest count is observed at the age benchmark of 25, with 175 occurrences, while the lowest count is observed for ages 65 and 70, with 5 occurrences each.

Box plot of customer's age



The box plot depicts that the 25th percentile falls at an age of 23, while the 75th percentile is at the age of 32.

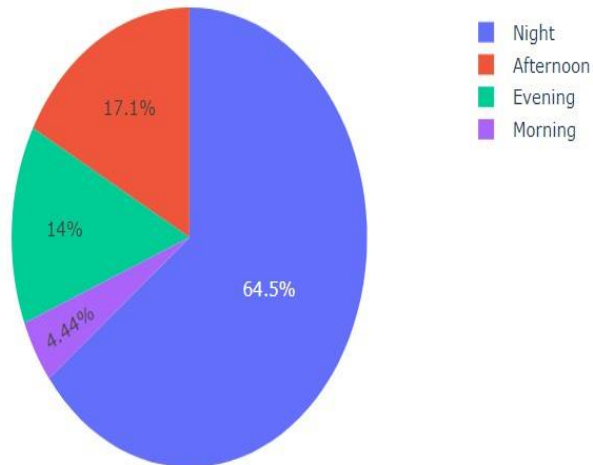
Sales distribution by age category



Roughly half of the sales distribution is accounted for by young adults, while the least sales are observed in the senior age group. Additionally, the adults' group has the second-largest share in sales (36.2%).



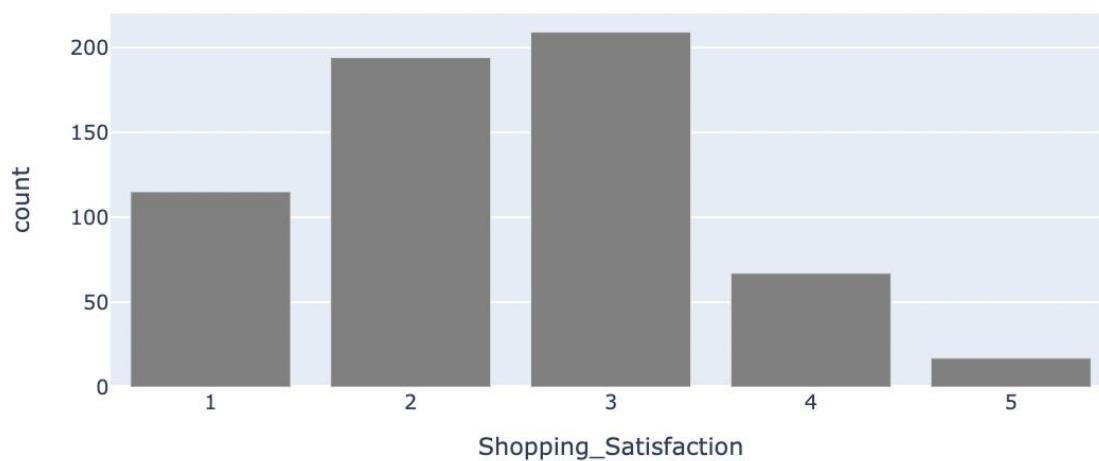
Analyze the time segments spent by Adult and Young Adult customers on A



Throughout the day, more than half of the time segment is depicted at night, while the morning accounts for less than 5%. However, the afternoon has the second-largest share, amounting to 17.1%.

Categorical rating of purchases

Shopping rating distribution : --



The highest benchmark of shopping rating is found to be "3" with a count of over 200, whereas it is as low as 20 for the rating of "5".



Distribution Plot of Shopping Satisfaction by Review Reliability :--

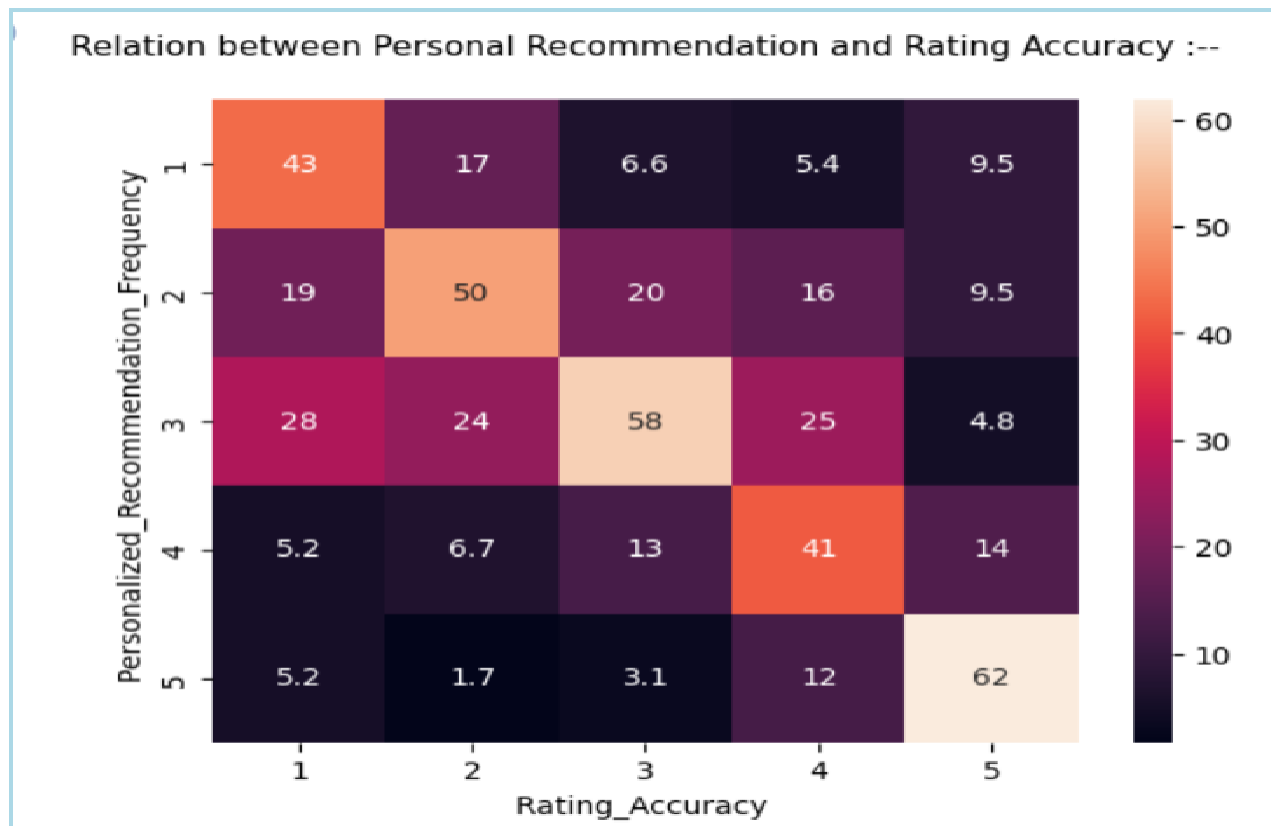


Customers who prioritize high customer review importance (rated 4 or 5) tend to experience better shopping satisfaction, typically ranging between a mode of 4 to 5. Conversely, those who assign lower importance to customer reviews (rated 1 or 2) exhibit lower shopping satisfaction, often falling within the range of 1 to 2 in terms of shopping satisfaction.

In summary, customer reviews serve as an excellent means to evaluate a product.

Heatmap

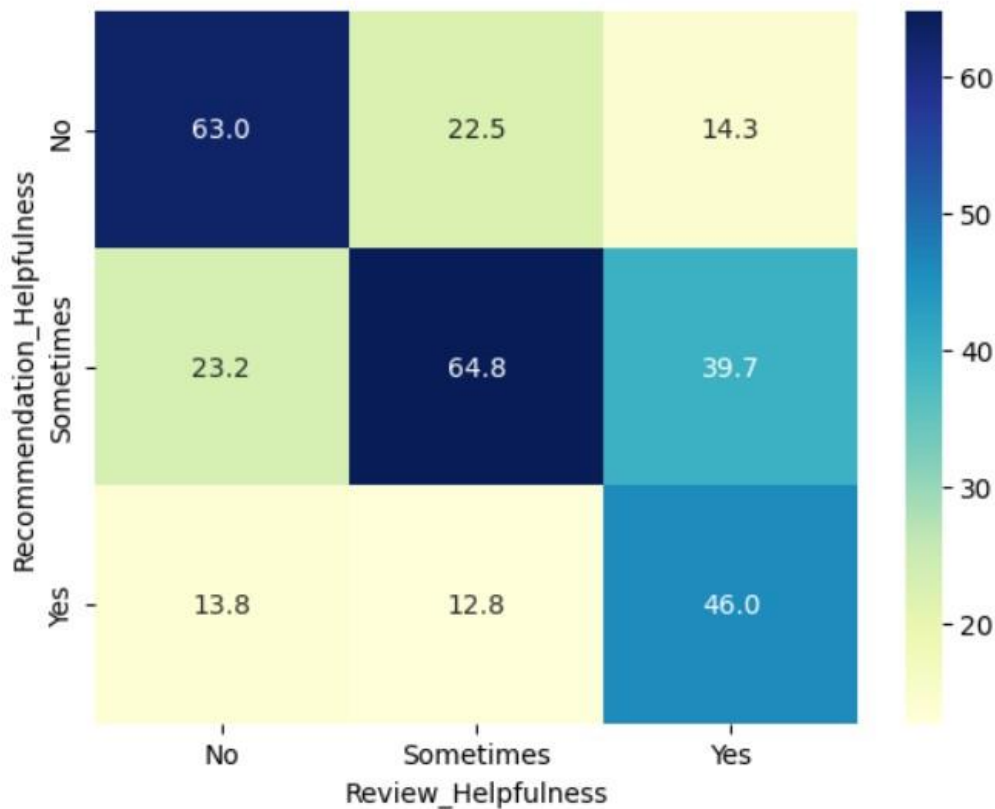
The heatmap highlights a strong correlation between Shopping Rating and Personalized Recommendation Frequency. Users receiving recommendations more frequently tend to experience better shopping outcomes. For instance: :--



- 47% of users receiving the most frequent recommendations (rated 5) report an excellent shopping experience (rated 5).
- 42% of users receiving frequent recommendations (rated 4) express having a good shopping experience (rated 4).
- 59% of users receiving moderate recommendations (rated 3) report a satisfactory shopping experience (rated 3).
- 46% of users receiving occasional recommendations (rated 2) indicate an unsatisfactory shopping experience (rated 2).
- 7% of users receiving rare recommendations (rated 1) note a very bad shopping experience (rated 1).

This observation underscores the positive impact of personalized recommendations on users' shopping experiences, with higher recommendation frequency aligning with higher satisfaction in shopping outcomes.

Relation between Recommendation Helpfulness and Review Helpfulness :--



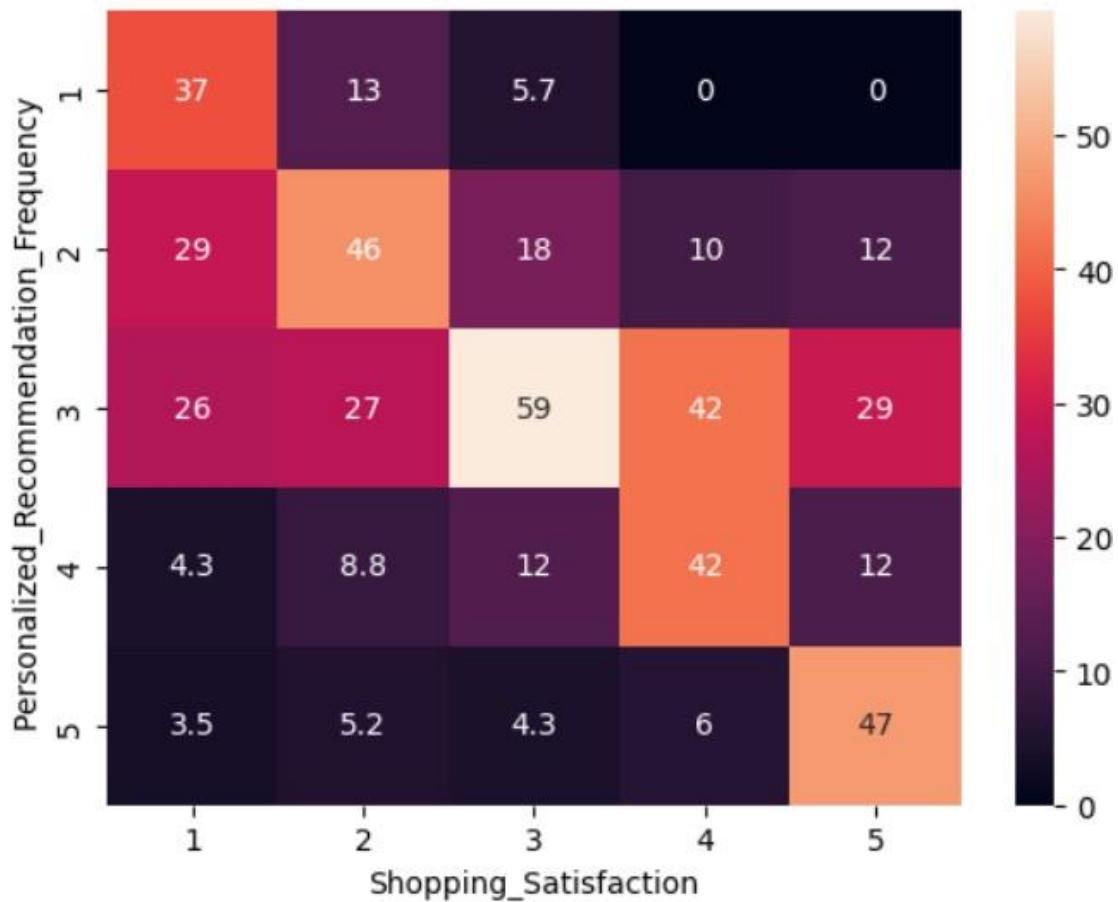
The above heatmap shows correlation between Review helpfulness and Recommendation helpfulness.

- 63% of users who never found recommendations helpful also never encountered useful reviews compared to 64.8% found recommendations helpful at times and also found reviews helpful occasionally.

- On the other hand, 46% of people who frequently found recommendations useful also reported experiencing better review satisfaction.

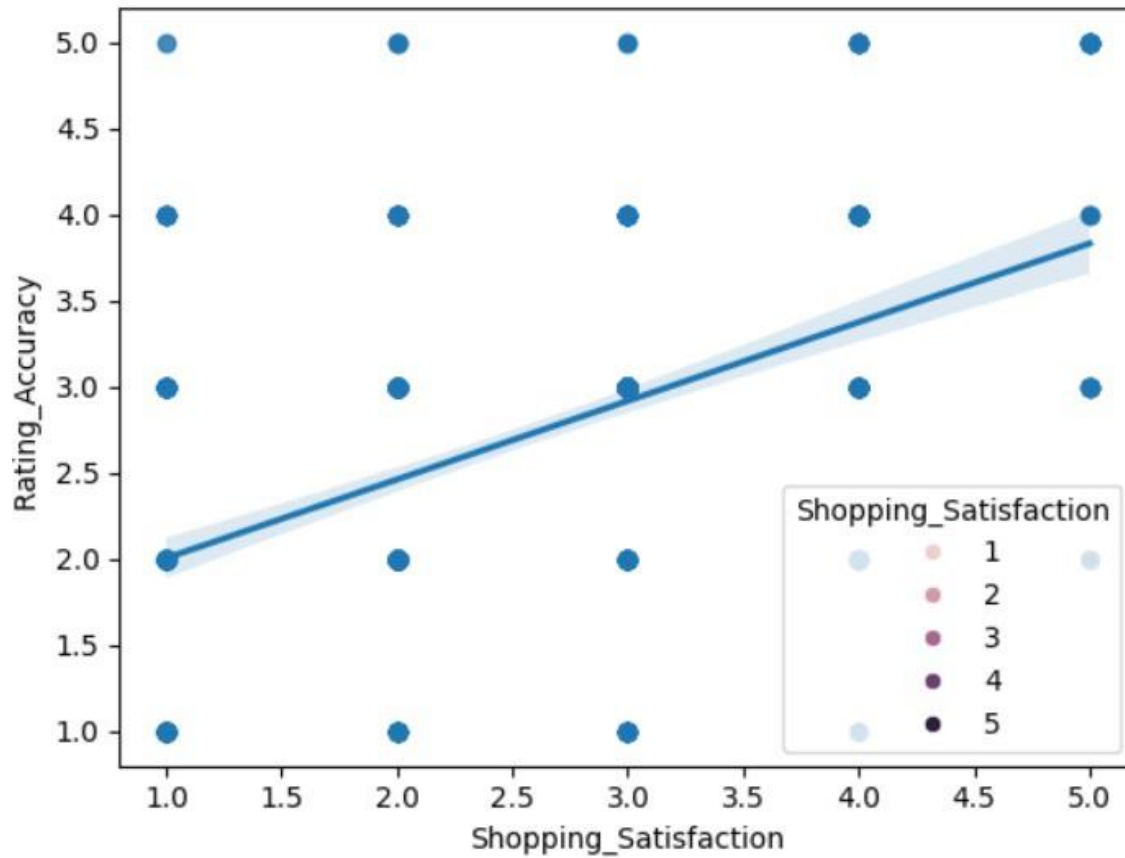
To conclude, there's a notable connection where users' perceptions of Recommendation helpfulness align somewhat with their experiences regarding Review helpfulness.

Relation between Personal Recommendation and Shopping rating :--

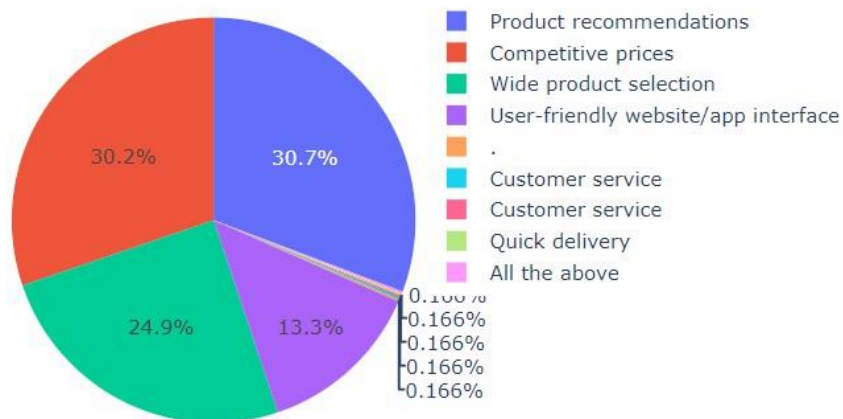


The heat map above reveals a positive correlation between Recommendation Frequency and Rating Accuracy. Higher accuracy ratings align with increased recommendation frequency. This indicates that products recommended based on accurate reviews provide customers with a reliable preview, highlighting the importance of accuracy in driving recommendations.

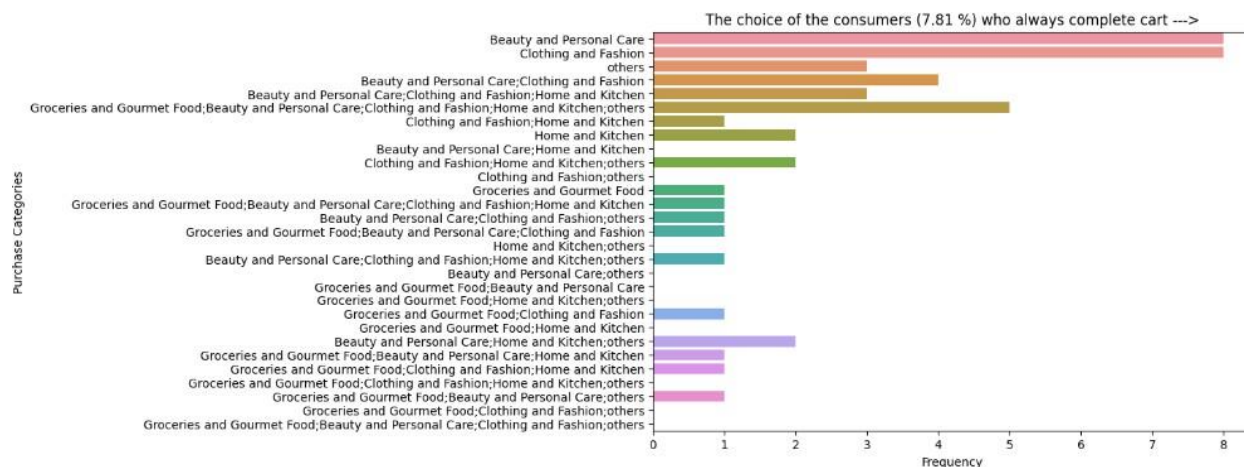
Correlation between Shopping rating and Rating Accuracy : 0.50



The correlation of 0.50 signifies a moderate positive link between Shopping Rating and Shopping Satisfaction, implying that as one increases, the other tends to rise as well, indicating a meaningful but not perfect association between the two factors.



Product recommendations, Competitive Pricing, and Wide Product Selection cover the major shares in the pie chart, at 30.7%, 30.2%, and 24.9%, respectively.



'Beauty and Personal Care' and 'Clothing and Fashion' exhibit the highest retention rates among the 7.81% of repeat customers who always complete their cart.

OLS Regression

We conducted an Ordinary Least Squares (OLS) regression to explore the relationship between various factors and customers' shopping satisfaction on Amazon to check if there is any relationship between these variables

Code input:

```
import pandas as pd
import statsmodels.api as sm

# Loading the dataset
df = pd.read_csv('Amazon Customer Behavior Survey.csv')

df['Purchase_Frequency'] = df['Purchase_Frequency'].map({'Few times a month': 1, 'Less than once a month': 2, 'Once a month': 3, 'Once a week': 4, 'Multiple times a week': 5})
df['Gender'] = df['Gender'].map({'Male': 1, 'Female': 2, 'Others': 4, 'Prefer not to say': 3})

string_to_int_map = {
    'Yes': 1,
    'No': 0
}

# Replacing non-numeric strings with their corresponding numeric values
df['Personalized_Recommendation_Frequency '] =
df['Personalized_Recommendation_Frequency '].replace(string_to_int_map)
df['Customer_Reviews_Importance'] =
df['Customer_Reviews_Importance'].replace(string_to_int_map)
df['Rating_Accuracy '] = df['Rating_Accuracy '].replace(string_to_int_map)
df['Customer_Reviews_Importance'] =
df['Customer_Reviews_Importance'].replace(string_to_int_map)

# Converting columns to integers
df['Personalized_Recommendation_Frequency'] =
df['Personalized_Recommendation_Frequency '].astype(int)
df['Customer_Reviews_Importance'] = df['Customer_Reviews_Importance'].astype(int)
df['Rating_Accuracy '] = df['Rating_Accuracy '].astype(int)
df['Customer_Reviews_Importance'] = df['Customer_Reviews_Importance'].astype(int)

# Selecting the dependent and independent variables
dependent_var = 'Shopping_Satisfaction'
```

```
independent_vars = ['age', 'Purchase_Frequency', 'Gender',
'Personalized_Recommendation_Frequency', 'Customer_Reviews_Importance',
'Rating_Accuracy ', 'Customer_Reviews_Importance']
```

```
X = sm.add_constant(df[independent_vars])
```

```
y = df[dependent_var]
```

```
# Fiting the OLS model
model = sm.OLS(y, X).fit()
```

```
# Printing out the statistics
model_summary = model.summary()
print(model_summary)
```

Output:-

Dep. Variable: Shopping_Satisfaction	R-squared: 0.366
Model: OLS	Adj. R-squared: 0.360
Method: Least Squares	F-statistic: 57.31
Date: Sat, 20 Apr 2024	Prob (F-statistic): 7.14e-56
Time: 23:02:41	Log-Likelihood: -723.69
No. Observations: 602	AIC: 1461.
Df Residuals: 595	BIC: 1492.
Df Model: 6	
Covariance Type: nonrobust	

	coefficient	std err	t	P> t	[0.025	0.975]
const	0.2688	0.185	1.456	0.146	-0.094	0.631
age	-0.0005	0.003	-0.154	0.878	-0.007	0.006
Purchase_Frequency	0.0062	0.024	0.257	0.798	-0.042	0.054
Gender	0.0639	0.047	1.366	0.173	-0.028	0.156
Personalized_Recommendation_Frequency	0.2031	0.036	5.594	0	0.132	0.274
Customer_Reviews_Importance	0.095	0.015	6.294	0	0.065	0.125

Rating_Accuracy	0.3924	0.042	9.385	0	0.475	0.475
Customer_Reviews_Importance	0.095	0.015	6.294	0	0.065	0.125

Omnibus: 7.369	Durbin-Watson: 1.671
Prob(Omnibus): 0.025	Jarque-Bera (JB): 10.645
Skew: -0.043	Prob(JB): 0.00488
Kurtosis: 3.646	Cond. No. 1.19e+16

Interpretation :-

Dependent Variable (Shopping Satisfaction):

The dependent variable in our model is “Shopping Satisfaction.” This represents customers’ satisfaction levels with their shopping experiences on Amazon.

Coefficients:

Constant (Intercept): When all other predictors are zero, the constant term (intercept) represents the predicted level of satisfaction from shopping. In this instance, it comes to around 0.2688.

Age: The age coefficient is extremely small (-0.0005), suggesting that age has no effect on the enjoyment of shopping.

Purchase Frequency: The coefficient of 0.0062 indicates a slight boost in shopping satisfaction with an increase in purchase frequency.

Gender: The coefficient, which is 0.0639, shows that, in comparison to being male (coded as 1), being female (coded as 2) is linked to somewhat greater levels of shopping satisfaction.

Personalized Recommendation Frequency: A higher frequency of personalized recommendations has a positive impact on purchasing satisfaction (coefficient = 0.2031).

Customer Reviews Importance: Customers who place a high value on reviews (coefficient = 0.0950) typically report happier buying experiences.

Significance:

Each coefficient's p-value provides information about its statistical significance.

Significant variables are those with p-values less than 0.05.

"Customer Reviews Importance," "Personalized Recommendation Frequency," and "Rating Accuracy" are statistically significant in our situation.

R-squared (R^2):

With an R-squared of 0.366, the model is able to account for about 36.6% of the variation in shopping pleasure.

Adjusted R-squared:

The number of predictors is taken into consideration by the adjusted R-squared (0.360). It penalizes the addition of unneeded variables.

Residuals:

The results of the Omnibus test indicate that the residuals are not normally distributed (p-value = 0.025).

Minimal autocorrelation in residuals is indicated by the Durbin-Watson statistic (1.671).

Interpretation:

Clients who obtain advice that are more tailored to them are typically happier.

Consumer reviews have a beneficial impact on consumer satisfaction.

Product satisfaction is greatly increased by accurate ratings.

Advanced Analysis

ANOVA (Analysis of Variance)

The code used is for feature selection using the SelectKBest method from the scikit-learn library in Python.

Code input:

```
import numpy as np
from sklearn.feature_selection import SelectKBest, f_classif
selector=SelectKBest(score_func=f_classif, k=10)
X_selected = selector.fit_transform(X,y)

# Get the selected feature indices
feature_indices = selector.get_support(indices=True)

# Get the names of the selected features
selected_features = X.columns[feature_indices]

# Create a new DataFrame with only the selected features
selected_data = df[selected_features]
# Print the selected features
print(selected_features)
```

Output:

```
Index(['Recomendation_Purchase_Status', 'Search_Result_Exploration',
      'Add_to_Cart_Browsing', 'Cart_Completion_Frequency',
      'Saveforlater_Frequency', 'Review_Reliability', 'Review_Helpfulness',
      'Recomendation_Purchase_Status ', 'Rating_Accuracy ',
      'Shopping_Satisfaction'],
      dtype='object')
```

Code input:

```
xnew=df.drop(['age', 'Gender','Purchase_Frequency', 'Purchase_Categories',
'Browsing_Frequency','Product_Search_Method',
'Customer_Reviews_Importance','Cart_Abandonment_Factors',
'Saveforlater_Frequency','Recommendation_Helpfulness',"Service_Appreciation","Improvement_Areas"],axis=1)
```

```

from imblearn.over_sampling import RandomOverSampler
sm=RandomOverSampler()
x,y=sm.fit_resample(xnew,y)

#minmaxscaler
ms=MinMaxScaler()
Xsc=ms.fit_transform(x)
Xsc

```

Output:

```

array([[1. , 1. , 1. , ..., 0.25, 0. , 0. ],
       [1. , 1. , 1. , ..., 0.25, 0.5 , 0.25],
       [0. , 1. , 1. , ..., 0.75, 0.5 , 0.5 ],
       ...,
       [1. , 1. , 1. , ..., 0.25, 0.5 , 0.75],
       [0.5 , 1. , 0.5 , ..., 0. , 0.25, 0. ],
       [0. , 1. , 1. , ..., 0.75, 0.5 , 0.75]])

```

```

#Splitting to test and train data
X_train,X_test,y_train,y_test=train_test_split(Xsc,y,test_size=0.1,random_state=4)

```

The above approach prepares the data effectively for training machine learning models, addressing common challenges such as class imbalance and feature scaling, and sets the stage for building robust and accurate predictive models.

Decision Tree

Code input:

```

dt=DecisionTreeClassifier(criterion='entropy',random_state=2,max_depth=10)
dt.fit(X_train,y_train)
y_pred3=dt.predict(X_test)
y_pred3

```

Output:

```

array([0, 0, 0, 0, 2, 2, 1, 0, 0, 0, 0, 2, 0, 2, 2, 2, 0, 0, 0, 2, 0, 2,
       2, 0, 0, 1, 2, 1, 1, 2, 2, 2, 0, 0, 1, 2, 2, 1, 2, 2, 0, 1, 2, 1,
       0, 2, 0, 2, 1, 0, 2, 1, 1, 2, 1, 0, 1, 0, 0, 1, 2, 1, 0, 0, 2, 1,
       2, 0, 2, 0, 0, 0, 2, 0, 0, 2, 0, 0, 1, 2, 1, 1])

```

Code input:

```
print(classification_report(y_test,y_pred3))
```

Output:

	Precision	Recall	f1-score	Support
0	0.53	0.9	0.67	20
1	0.74	0.56	0.64	25
2	0.83	0.65	0.73	37
accuracy			0.68	82
macro avg	0.70	0.70	0.68	82
weighted avg	0.73	0.68	0.68	82

Overall accuracy is 0.68.

Random Forest Classifier

Code input:

```
rf=RandomForestClassifier(criterion= 'entropy', max_depth= None, min_samples_leaf= 1,  
min_samples_split= 4,n_estimators= 200)  
rf.fit(X_train,y_train)  
y_pred4=rf.predict(X_test)  
y_pred4
```

Output:

```
array([0, 0, 0, 0, 2, 0, 1, 0, 0, 0, 0, 2, 0, 1, 2, 2, 0, 0, 0, 1, 0, 2,  
       2, 0, 0, 1, 2, 1, 0, 2, 2, 2, 0, 0, 1, 2, 2, 1, 2, 2, 0, 1, 2, 1,  
       2, 2, 0, 2, 1, 0, 2, 1, 1, 2, 1, 0, 1, 0, 0, 1, 2, 1, 0, 2, 2, 1,  
       2, 1, 2, 0, 0, 0, 2, 0, 2, 2, 0, 0, 2, 2, 1, 1])
```

Code input:

```
print(classification_report(y_test,y_pred4))
```

Output:

	Precision	Recall	f1-score	Support
0	0.59	0.95	0.73	20
1	0.80	0.64	0.71	25

2	0.87	0.70	0.78	37
accuracy			0.74	82
macro avg	0.75	0.76	0.74	82
weighted avg	0.78	0.74	0.75	82

Overall accuracy is 0.74.

→ Interpreting the outputs and comparing the performance of the Decision Tree Classifier and the Random Forest Classifier.

- **Precision:** Random Forest identifies true positives more accurately for all classes.
- **Recall:** Random Forest captures a higher proportion of actual positives, minimizing false negatives.
- **F1-score:** By balancing precision and recall, the F1-score confirms Random Forest's overall superiority.
- **Accuracy:** Random Forest makes fewer errors, resulting in a higher overall accuracy.
- **Macro/Weighted Average:** Both metrics, accounting for class imbalance if present, favor the Random Forest Classifier.

In summary, based on these metrics, the Random Forest Classifier seems to be a better choice for this classification task compared to the Decision Tree Classifier. It shows improvements in precision, recall, F1-score, and overall accuracy.

Conclusion

Business Implications :-

Recognizing the Demographics and Behavior of Customers:

Companies can adjust their product offerings and marketing methods to target the majority of their client base, in this case, women. Targeting young adults and adults with high levels of purchasing activity is another option. Businesses that identify the evening as the ideal time to shop might modify their advertising and promotions to draw in customers during these hours.

Product Categories and Customer Preferences:

Companies can focus their marketing and product development efforts and resources on high-performing categories such as apparel, fashion, beauty, and personal care. Businesses can tailor their product offerings and marketing messaging to more effectively connect with their target audience by taking gender-based preferences into account.

Shopping Satisfaction Factors:

Companies can make investments to improve these areas after realizing the substantial influence that user evaluations and tailored recommendations have on customers' shopping satisfaction. This could entail offering individualized purchasing experiences, improving recommendation algorithms, and requesting and encouraging good ratings.

Problems and Opportunities for Improvement:

Companies can close the relevance gap by optimizing recommendation algorithms to better meet the demands of their clients right away. Providing superior customer service, superior products, and increased choice autonomy to customers can improve their entire shopping experiences and satisfaction levels.

Opportunities for Improvement:

Businesses can increase customer happiness and loyalty by concentrating on enhancing customer service, product quality, and environmental initiatives (such as reducing package waste). Higher conversion rates and more customer engagement might result from recommendations that are more closely aligned with current requirements and preferences.

Relationship Between Suggestions and Reviews:

Maintaining accurate and dependable review systems is crucial to generating successful recommendations, as evidenced by the positive association found between helpful recommendations and reviews' perceived usefulness.

Limitations :-

Small Sample Size: Only 13 days' worth of records are included in the report, which is a rather short time span. There is less chance of formulating predictions or reaching firm conclusions because of this tiny sample size.

Incomplete Data: People who did not purchase any things are not included in the dataset; only records for those who made purchases are included. This restriction could generate bias and have an impact on how broadly the results can be applied.

Existence of Underage Customers: The collection contains records pertaining to customers who are younger than eighteen years old. Without particular circumstances—which the data does not address—this seems improbable. This disparity calls into doubt the data's completeness and veracity.

Future Directions :-

Based on an analysis of customer demographics, behavior, product preferences, and satisfaction criteria, the following future directions have been suggested:

- Examine the actions of consumers who are not making purchases to learn about their browsing preferences and motivations.
- By extending the dataset to encompass a longer time span, you can perform long-term trends analysis.
- Determine areas where the shopping experience needs to be improved by tracking client satisfaction over time.
- Utilizing past purchase data, forecast future demand for certain products or product groups.
- To determine distinct client segments and adjust marketing strategy accordingly, conduct segmentation analysis.
- To obtain a thorough grasp of client behavior and preferences, integrate other data sources like social media, website analytics, and customer feedback surveys.
- These upcoming paths seek to use data-driven decision-making to improve comprehension of consumer behavior, streamline operations, and spur expansion in the e-commerce sector.

Citations

Anova (Analysis of Variance)

<https://www.geeksforgeeks.org/anova-formula/>

History of Ecommerce

<https://blog.miva.com/the-history-of-ecommerce-how-did-it-all-begin>

About Amazon

[https://en.wikipedia.org/wiki/Amazon_\(company\)](https://en.wikipedia.org/wiki/Amazon_(company))

Dataset Link

<https://www.kaggle.com/>

Decision Tree Classification

<https://www.youtube.com/watch?v=ZVR2Way4nwQ>

Random Forest Classifier

<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>